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**AdaptEd: Harnessing AI for Personalised and Multimodal
Learning in Higher Education**

SUPERVISOR

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**I declare that all work submitted for this coursework is the work of Ganiyat Adekunle, S2110996,
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Abstract

Delivering personalised and engaging learning experiences remains a challenge as institutions struggle to meet the diverse needs of students. The AdaptEd project aims to develop an AI-powered Learning Management System (LMS) plugin to address these challenges. By integrating natural language processing, machine learning, and multimedia content generation, AdaptEd provides multimodal content, personalised content recommendations, and proactive support for students. The literature and technology review span diverse learning needs, traditional educational limitations, and AI applications and technologies in personalised learning. A mixed-methods evaluation will combine quantitative performance metrics with qualitative feedback to assess its effectiveness. Expected outcomes include improved student engagement, higher retention rates, and enhanced academic performance.

Keywords: AI in education, learning styles, personalised learning, multimodal learning, learning analytics, recommender systems, machine learning

1.0 Introduction

1.1 Project Background

Higher education faces significant challenges in delivering effective learning experiences tailored to diverse student needs, preferences, and abilities. Traditional one-size-fits-all teaching methods result in disengagement and suboptimal outcomes, with up to 40% of students feeling disconnected from their coursework (Popenici & Kerr, 2017). Existing Learning Management Systems (LMS) lack advanced personalisation capabilities, placing a significant burden on educators who spend over 10 hours weekly on content customisation and student support (Attri et al., 2023). Moreover, these systems often fail to proactively identify and support at-risk students, contributing to higher dropout rates, with about 30% of undergraduates leaving within their first year (NCES, 2020).

The goal of this project is to address these critical issues by developing AdaptEd, an AI-powered LMS plugin designed to provide personalised content recommendations, multimodal content delivery, and proactive student support. Utilising advanced technologies such as natural language processing and machine learning, AdaptEd dynamically adapts educational content to meet individual student needs. It also includes a recommender system and learning analytics service to identify at-risk students and offer actionable insights for educators. By integrating these technologies, AdaptEd aims to enhance student engagement, comprehension, and academic success, filling the gaps in current educational models (Holmes et al., 2019; Luckin et al., 2016).

1.2 Project Aims and Objectives

1.2.1 Problem Statement

The current landscape of higher education faces significant challenges in delivering effective and engaging learning experiences that cater to the diverse needs, preferences, cognitive abilities, accessibility requirements, and learning styles of students. Traditional one-size-fits-all approaches to teaching and learning often fail to address the individual differences among learners, leading to disengagement, suboptimal learning outcomes, and an increased risk of academic failure. While some adaptive learning technologies exist, they are frequently fragmented and not seamlessly integrated with popular LMS platforms. They require significant manual effort from educators and fail to provide a comprehensive, AI-driven personalised learning experience, limiting their effectiveness and scalability. The lack of personalised support and timely intervention for students who may be struggling or at risk of falling behind further exacerbates these challenges.

Existing learning management systems (LMS) and educational technologies, while powerful in their own right, often lack the advanced capabilities necessary to deliver truly personalised and adaptive learning experiences. These systems often rely on manual intervention from educators to tailor content and provide individualised support, which can be time-consuming, resource-intensive, and difficult to scale. The absence of sophisticated learning analytics and predictive modelling tools within these systems hinders the ability to identify and proactively support students who may be at risk of academic failure. To address these challenges, there is a critical need for an intelligent, AI-powered LMS plugin that can seamlessly integrate with existing platforms and deliver highly personalised and engaging learning experiences.

The proposed AdaptEd plugin aims to bridge this gap by utilising AI technologies for personalised content recommendations, multimodal content delivery, and proactive student support. AdaptEd will leverage technologies such as natural language processing, machine learning, and multimedia.

generation to dynamically adapt educational content based on individual student needs and preferences. Additionally, it will feature a recommender system for tailored learning materials and a learning analytics service to identify at-risk students and provide actionable insights for educators.

1.2.2 Secondary Research Objectives

1. Examine the cognitive and affective factors influencing learning styles and preferences, using insights from educational psychology, cognitive science, and cross-cultural studies.
2. Critique traditional educational models to identify limitations and the need for adaptive approaches through a review of historical and contemporary practices.
3. Analyse the impact of personalised and adaptive learning on educational equity and academic performance.
4. Investigate the evolution and integration of AI technologies in personalised learning, focusing on multimodal content, educational recommender systems, and learning analytics through literature and case studies.
5. Conduct a comparative analysis of personalised and adaptive learning platforms to evaluate their strengths, limitations, and effectiveness in improving student engagement and performance.
6. Examine the role of instructional design principles, multimedia learning theories, and user-centred design in developing engaging and personalised learning experiences.
7. Investigate challenges in integrating AI-powered solutions with existing LMS, ensuring interoperability, data privacy, and accessibility.
8. Examine frameworks and metrics for evaluating personalised learning systems.
9. Investigate the ethical and privacy implications of using student data in personalised learning, recommender systems, and learning analytics, examining relevant regulations and best practices in data governance and privacy protection.

1.2.3 Primary Research Objectives

1. Develop an intelligent and context-aware recommender system to suggest educational resources based on learning preferences and similarities with other learners, using advanced filtering techniques and evaluating its effectiveness through experimentation and user studies.
2. Design and develop a multimodal content generator that dynamically converts text-based educational content to audio, slides, and video using machine learning and multimedia generation techniques while adhering to instructional design principles.
3. Develop an AI-driven virtual assistant to provide personalised guidance, support, and feedback to students using natural language processing and conversational AI technologies.
4. Construct a learning analytics service to monitor and analyse student engagement, performance, and progress, using machine learning and statistical techniques to identify patterns, predict at-risk students, and provide actionable insights while ensuring data privacy and ethical considerations.
5. Integrate AdaptEd with popular LMS, ensuring interoperability, performance, and scalability through industry-standard protocols and APIs, with rigorous testing to ensure compatibility and a cohesive user experience.
6. Evaluate AdaptEd through a mixed-methods approach, combining quantitative analysis of student performance and engagement metrics with qualitative insights from user studies, focus groups, and educator feedback to assess its effectiveness in improving personalised learning and supporting at-risk students.

2.0 Literature and Technology Review

2.1 Literature review

2.1.1 The diverse learning needs and preferences of students

Understanding the diverse learning needs and preferences of students is crucial for effective education. Learning styles, such as visual, auditory, and kinesthetics' modalities, significantly impact student performance. Research into learning styles, such as VARK (Visual, Auditory, Reading/Writing, Kinesthetics), and Kolb's Experiential Learning Theory highlights the importance of tailoring educational strategies to different preferences and cognitive abilities (Kolb, 2014). The VARK model categorises learners to help educators develop strategies that engage students in the ways they learn best (Fleming and Mills, 1992). Figure 1 demonstrates the VARK model.

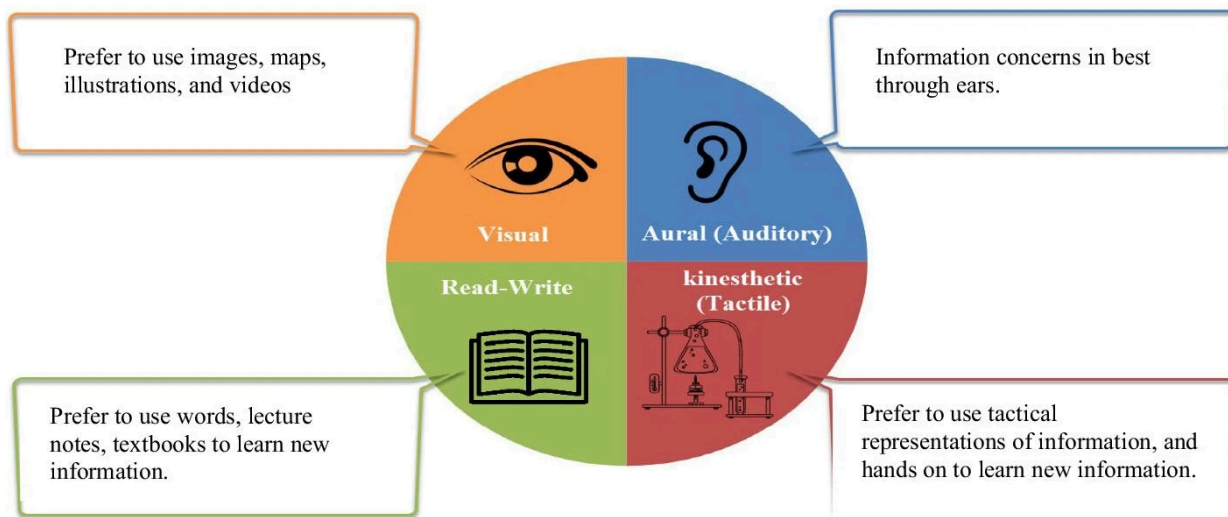


Figure 1: VARK model (El-Sabagh, 2021, Fig. 1.)

Research by Rahman and Budivanto (2019) demonstrates that adaptive e-learning systems, which align educational content with these styles, enhance student engagement and knowledge retention. However, the validity of learning styles as a concept is contested. An and Carr (2017) argue that tailoring education to specific learning styles lacks empirical support and suggest focusing on cognitive and metacognitive strategies instead. Cognitive abilities and processing speeds further differentiate learners. For instance, students with higher cognitive abilities process information more quickly, necessitating advanced content, while those with slower processing speeds benefit from additional support. Accessibility requirements also vary, with some students needing assistive technologies or adapted materials to fully participate in learning activities. The difficulty in early identification of at-risk students compounds these challenges as educators struggle to discern and address individual support needs (Khamparia & Pandey, 2020). This gap emphasises the need for personalised and adaptive learning systems that can dynamically cater to diverse student profiles, thereby enhancing engagement and reducing the risk of academic failure. Critiquing traditional educational models, which often fail to provide personalised support and timely interventions, sets the stage for addressing these varied needs.

2.1.2 Traditional educational approaches and their limitations

Historically, educational practices such as lecture-based instruction, standardised testing, and fixed curricula prioritised uniformity and standardisation, aiming to streamline teaching processes. Although education systems have evolved from rigid, standardised methods to more flexible and student-centred

approaches, many traditional models still lack the adaptability required to cater to individual learning preferences and cognitive abilities (Attri et al., 2023). This rigidity frequently leads to disengagement and suboptimal learning outcomes among students whose needs are not met by conventional methods (Hashim et al., 2022). Lecture-based instruction often fails to engage students, while standardised testing assesses all students using the same metrics, ignoring individual strengths and weaknesses. Additionally, traditional methods struggle to identify and support at-risk students promptly, exacerbating challenges related to academic performance and retention (Albreiki et al., 2023). The lack of sophisticated learning analytics and predictive modelling tools further hampers the ability to provide timely and effective interventions, highlighting the need for more advanced and integrated solutions (Adiguzel, Kaya, & Cansu, 2023). Also, current methods often rely heavily on manual efforts to tailor support, which is resource-intensive and not scalable. This inadequacy points to a pressing need for adaptive learning technologies that can offer personalised and responsive educational experiences, seamlessly integrating with existing LMS platforms.

2.1.3 The benefits and impact of personalised and multimodal learning approaches

Personalised and multimodal learning approaches, leveraging technologies like intelligent tutoring and recommender systems, provide tailored experiences (Zawacki-Richter et al., 2019). They integrate visual, auditory, and kinesthetics materials, catering to diverse learning preferences and enhancing retention (Blikstein, 2013). Studies suggest that adaptive learning systems significantly improve outcomes by accommodating different learning styles and abilities (Graf et al., 2007). Real-time analytics in personalised systems enable prompt intervention for at-risk students, contrasting with reactive traditional models (Al-Shabandar et al., 2019). These approaches enhance engagement and academic success and promote inclusive education, showcasing the transformative potential of AI integration in education.

2.1.4 The application of AI technologies in creating personalised and adaptive learning experiences

2.1.4.1 Evolution and the use of AI in Education

The evolution and integration of AI technologies in education have the potential to revolutionise personalised and adaptive learning experiences. As AI technologies continue to evolve, they can address the limitations of traditional educational models and create more inclusive and effective learning environments. AI in Education (AIEd) leverages advanced algorithms to adapt instructional content, identify at-risk students, and provide real-time feedback, making learning more responsive and tailored to individual needs. AI applications in education have evolved from simple automated grading systems to sophisticated platforms that offer personalised learning experiences. Early implementations focused on automating administrative tasks, such as grading and attendance, to reduce the burden on educators. However, recent advancements have enabled the development of intelligent tutoring systems (ITS), adaptive learning platforms, and learning analytics tools that provide more nuanced and personalised educational experiences (Holmes et al., 2019).

2.1.4.2 AI Technologies

Multimodal Content Generation

AI significantly advances multimodal content generation by integrating text, audio, video, and interactive elements to accommodate diverse learning preferences. Natural Language Processing (NLP) generates text summaries, explanations, and interactive dialogue-based materials, while machine learning algorithms analyse large datasets to personalise content for individual student needs (Zawacki-Richter et al., 2019). This is particularly impactful in STEM fields, where AI-generated

simulations and interactive labs provide hands-on experience and visualisations of complex concepts (Kaplan & Haenlein, 2019).

Educational Recommender Systems

AI enhances education through recommender systems, which utilise machine learning to suggest personalised learning materials, resources, and activities. These systems consider past performance, learning pace, and specific difficulties to tailor suggestions, thereby improving learning efficiency and engagement (Manouselis et al., 2011). They save educators time by automating the identification of suitable materials, allowing more focus on instruction, and enabling students to discover new resources and learning paths (Santos & Boticario, 2015).

Learning Analytics and Identifying At-Risk Students

AI plays a crucial role in learning analytics, particularly in identifying at-risk students. Learning analytics involves collecting, analysing, and reporting data to optimise learning environments. AI-powered predictive modelling and early warning systems detect patterns indicating academic struggles, such as low engagement or poor performance (Siemens & Long, 2011). Early identification facilitates targeted interventions, like tutoring sessions, to prevent academic failure, contrasting with traditional reactive methods where interventions often occur after students have already fallen behind (Arnold & Pistilli, 2012).

AI-Driven Virtual Assistants in Education

AI-driven virtual assistants, such as chatbots, provide real-time support and enhance learning experiences using NLP to respond to student queries, offer explanations, and guide learners through content. Available 24/7, these assistants offer support outside regular hours and reduce educators' workloads by handling routine queries and providing immediate feedback (Vázquez-Cano et al., 2021; Kiptonui, 2013).

2.1.4.3 Challenges and Considerations

The integration of AI in education offers significant potential benefits but also presents several challenges and considerations. Ensuring the ethical use of AI, particularly regarding data privacy and security, is a major challenge, necessitating compliance with legal standards like the GDPR and adherence to ethical guidelines to protect student privacy (Popenici & Kerr, 2017). Algorithmic bias is also a concern, as AI systems can reinforce existing biases if trained on unrepresentative datasets, highlighting the need for diverse and comprehensive data to ensure fair educational outcomes (Holmes et al., 2019). Additionally, the adoption of AI requires substantial investment in infrastructure and professional development, with educators needing training in both technical skills and the ethical and pedagogical implications of AI in education (Luckin et al., 2016).

2.1.5 Case studies and evaluations of existing adaptive learning platforms and AI-powered educational solutions

The transformative potential of AI in education is evidenced by numerous case studies. The ASSISTments platform, for instance, provides personalised feedback for math problems, effectively addressing individual learning gaps and significantly enhancing student performance (Heffernan & Heffernan, 2014). Similarly, OpenStax Tutor leverages AI and open educational resources to create customised learning experiences, resulting in increased student engagement and improved academic outcomes (OpenStax, 2020).

Carnegie Learning's MATHia software exemplifies AI-driven adaptive learning, creating individualised learning paths. A study involving over 1,000 middle school students revealed that those using MATHia

scored 20% higher on standardised math tests compared to their peers using traditional methods (Ritter et al., 2007). Despite these benefits, issues of algorithmic fairness and potential biases in the training data pose significant challenges.

Rahman and Budivanto (2019) demonstrated an adaptive e-learning system tailored to visual, auditory, and kinesthetics learning styles. Their implementation for 150 students resulted in a 30% increase in engagement and a 25% improvement in knowledge retention. However, the system's reliance on accurate initial learning style assessments can lead to suboptimal content delivery if misidentified. Khamparia and Pandey (2020) reviewed various adaptive e-learning systems and reported a 40% reduction in dropout rates and a 35% increase in engagement. Despite these advantages, they highlighted the complexities and costs associated with implementing such systems, which can pose financial and technical barriers to widespread adoption.

Graf et al. (2007) explored the Felder-Silverman learning style dimensions in adaptive learning systems. Their analysis of 200 students showed a 20% improvement in learning outcomes and a 15% increase in satisfaction. However, they raised concerns about privacy due to continuous monitoring and data collection, highlighting the need for robust data security and student consent.

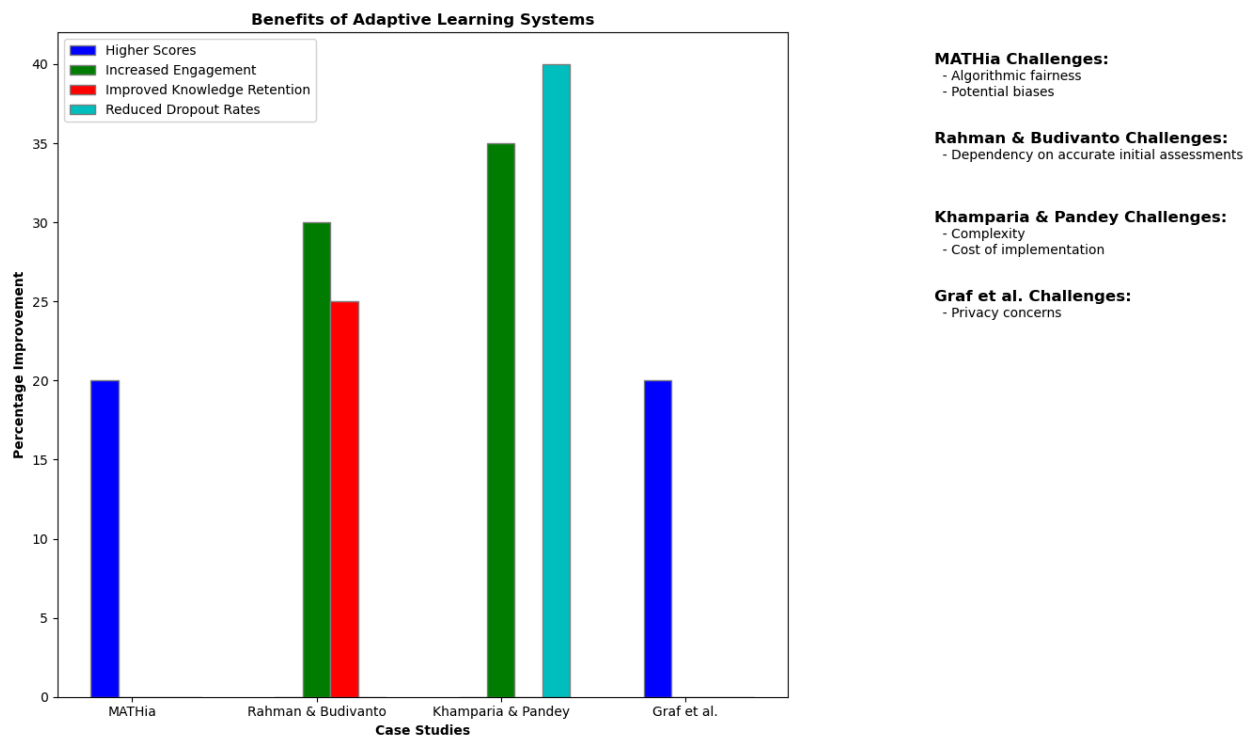


Figure 2: Benefits of Adaptive Learning Systems

These case studies illustrate the significant benefits of adaptive learning platforms for enhancing engagement and academic performance through personalised experiences. Nonetheless, they also highlight critical challenges, including the need for accurate assessments, financial and technical barriers, and privacy concerns. Addressing these issues is essential to fully realising the potential of AI-powered adaptive learning technologies in education (Popenici & Kerr, 2017).

2.1.6 Best practices and guidelines for designing effective multimodal learning content

Designing effective multimodal learning content requires adherence to instructional design principles, cognitive load management, and accessibility considerations. Instructional design principles, such as

those outlined by the Universal Design for Learning (UDL) framework, emphasise the need for flexible and inclusive educational materials that accommodate diverse learning needs (Bontchev & Vassilev, 2012). Cognitive load management involves structuring content to avoid overwhelming students with excessive information and ensuring that learning materials are presented in a clear and manageable format (Hamal et al., 2022).

Accessibility considerations are also crucial in creating inclusive educational environments. This includes designing content that is accessible to students with disabilities, using assistive technologies, and ensuring compliance with accessibility standards (Boesl, Achtenberg, & Bergler, 2023). By incorporating these best practices, educators can create multimodal learning experiences that are engaging, effective, and inclusive.

2.1.7 Challenges and considerations in integrating AI-powered solutions with Learning Management Systems (LMS)

Integrating AI-powered solutions with learning management systems (LMS) poses several technical, ethical, and privacy challenges. Ensuring seamless interoperability between AI tools and LMS platforms requires adherence to technical standards and protocols like Learning Tools Interoperability (LTI) and Experience API (xAPI) (Santos & Boticario, 2015). These standards facilitate data exchange and integration, ensuring that various tools can work together smoothly within an LMS environment. Privacy and data security are crucial, as AI usage raises concerns about consent, data protection, and ethical use, necessitating compliance with regulations such as GDPR (Popenici & Kerr, 2017). Transparency in data usage and AI decision-making is vital for trust. Additionally, effective user adoption requires training in technical skills and ethical implications, and overcoming resistance to AI-enhanced teaching methods requires robust support and ongoing professional development for educators (Holmes et al., 2019).

2.1.8 Evaluation and Assessment of Personalised Learning Systems

Effective evaluation and assessment of personalised learning are crucial to understanding its impact on student engagement, comprehension, and overall academic performance. Various methods and metrics can be employed to evaluate the effectiveness of personalised learning systems.

Methods for Evaluation

Table 1: Methods for evaluating personalised learning systems

Methods	Description
Learning Analytics Frameworks	Frameworks such as the Learning Analytics Cycle, which includes steps like data collection, analysis, action, and reflection, provide a structured approach to evaluating personalised learning (Clow, 2012).
Randomised Controlled Trials (RCTs)	RCTs are considered the gold standard for evaluating the impact of educational interventions. By randomly assigning students to either a personalised learning group or a control group, researchers can assess the effectiveness of personalised learning in a controlled environment (Koedinger et al., 2013).
Quasi-Experimental Designs	When RCTs are not feasible, quasi-experimental designs can be used. These designs involve comparing groups that are not

	randomly assigned but are matched on relevant characteristics to isolate the effects of personalised learning interventions (Cheung & Slavin, 2013).
Comparative Analysis	Comparing personalised learning systems with traditional educational approaches can highlight strengths and areas for improvement. Studies have shown that personalised learning can lead to higher engagement and better academic outcomes compared to traditional methods (Pane et al., 2017).

Metrics for Student Engagement, Comprehension, and Performance

Table 2: Metrics for evaluating personalised learning systems

Metrics	Description
Engagement Metrics	Metrics such as time spent on tasks, participation in discussions, and frequency of interactions with learning materials can indicate levels of student engagement. Learning management systems can track these metrics automatically, providing valuable data for analysis (Baker & Inventado, 2014).
Comprehension Metrics	Assessing comprehension often involves analysing student performance on formative and summative assessments. Metrics such as quiz scores, assignment grades, and performance on standardised tests can provide insights into how well students understand the material (Shute & Wang, 2016).
Performance Metrics	Overall academic performance can be measured using grades, retention rates, and completion rates. Additionally, comparing the performance of students in personalised learning environments with those in traditional settings can highlight the effectiveness of personalised approaches (Holmes et al., 2019).

2.1.9 Research Gaps and Future Directions

Identifying gaps and limitations in the current literature is critical for advancing the field of AI in education. Key areas for future research include the development of more robust and interpretable AI models, the integration of interdisciplinary perspectives, and the exploration of emerging technologies (Bozkurt et al., 2021). Additionally, there is a need for more comprehensive studies that address the ethical and practical challenges of implementing AI in educational settings (Hamal et al., 2022).

Future research should also focus on the potential of AI to enhance educational equity and access, particularly for underserved and marginalised populations (Attri et al., 2023). By addressing these gaps and exploring new directions, researchers and educators can continue to innovate and improve the use of AI in education, ultimately promoting better learning outcomes for all students.

2.2 Technology Review

2.2.1 Multimodal Content Generation Technologies

a) Text-to-Speech (TTS) Generation

Generating multimodal educational content enhances accessibility and caters to diverse learning preferences by offering content in formats such as audio, video, and slides. This section reviews different text-to-speech (TTS) synthesis methods.

Shen et al. (2018) introduced WaveNet, achieving high naturalness and audio quality, but it is computationally intensive and requires substantial training data. Google Text-to-Speech (gTTS) is a cloud-based solution offering high-quality, natural-sounding audio, multiple language support, and scalability with minimal computational resources, making it suitable for educational contexts. Sotelo et al. (2017) presented Char2Wav, an end-to-end neural network model that simplifies pre-processing but falls short in audio quality and control. Wang et al. (2017) used NLP techniques for text-to-audio and video summaries, producing high-quality outputs dependent on text complexity and the NLP model.

gTTS is the most suitable choice for AdaptEd's multimodal content generation due to its ease of implementation, high-quality output, language support, and scalability.

Table 3: Comparative analysis of TTS techniques

Source	Approach	Environment	Outcome	Strengths	Limitations
Shen et al. (2018)	WaveNet	Laboratory setting	High naturalness and audio quality	Captures complex speech characteristics	Computationally expensive
gTTS (Google Text-to-Speech)	Cloud-based TTS	Various educational contexts	High-quality, natural-sounding audio	Easy to implement, supports multiple languages	Requires internet connection
Sotelo et al. (2017)	Char2Wav	Laboratory setting	Simplified pipeline, end-to-end model	Eliminates the need for complex pre-processing	Limited audio quality and control
Wang, Yi, et al. (2020)	NLP techniques	Simulated educational setting	Efficiently converts text to audio and video summaries	High-quality audio and video generation	Quality depends on text complexity and the NLP model

b) Slide Generation

The studies explored various approaches for generating slides from textual content. Li et al. (2021) used unsupervised mutual learning to extract topic-relevant sentences, showing superior performance in maintaining topic relevance but limited to text extraction. Fu et al. (2022) applied a template-based NLP approach, providing straightforward slide generation but with limited flexibility and monotonous

results. Sun et al. (2021) used a hybrid of rule-based methods and machine learning to handle complex content, offering comprehensive summaries but requiring extensive rule tuning. The method adopted for AdaptEd is a combination of NLP models and a Python library using Hugging Face Transformers for text summarisation and python-pptx for slide creation. This approach is selected for its high accuracy in text summarisation and flexibility in slide creation.

Table 4: Comparative analysis of text-to-slide generation techniques

Sources	Methodology	Technology/Framework	Outcome	Strengths	Limitations
Li et al. (2021)	Unsupervised mutual learning	Neural networks, log-linear model	Topic-relevant sentence extraction	Topic-aware, effective without labelled data	Limited to text, manual annotation is needed for evaluation.
Fu et al. (2022)	Template-based NLP	NLP techniques	Summarise key points using templates	User-friendly, straightforward	Monotonous slides lack flexibility.
Sun et al. (2021)	Hybrid rule-based and machine learning	Rule-based methods, machine learning	Summarises main contributions, methodologies, and findings	Handles complex content, comprehensive summaries	Extensive tuning is needed; it depends on rule quality.
Hugging Face Transformers + python-pptx	Advanced NLP for summarisation, Python library for slide generation	Hugging Face Transformers, Python pptx	High-quality text summarisation and automated slide creation	High accuracy and relevance of summaries, flexible slide creation	Requires familiarity with advanced NLP and Python libraries

c) Video Generation

The comparative analysis of text-to-video generation approaches reveals distinct methodologies and technologies across different systems. CogVideo utilises multi-frame-rate hierarchical training, fine-tuning a pre-trained text-to-image model to enhance text-clip alignment and generation accuracy (Hong et al., 2022). C3LLM employs a multimodal LLM structure with hierarchical audio tokenisation for versatile content generation, although it faces complexity in training and real-time integration (Wang et al., 2024). Make-a-Video leverages GANs to produce high-quality, realistic video content but requires extensive computational resources and large datasets (Singer et al., 2022). Mo, Shi, and Tian (2024) focus on synchronising text-to-audio generation with video content, effectively enhancing audio-visual coherence, though real-time synchronisation for complex videos remains challenging. Each system's unique strengths and limitations highlight the ongoing need for improved efficiency, dataset availability, and real-time capabilities in text-to-video generation technology.

Mo, Shi, and Tian's (2024) approach will be adopted for the AdaptEd plugin to synchronise the audio output and the slide output as shown in Figure 3 because of its simplicity and effective audio-visual synchronisation.

Table 5: Comparative analysis of text-to-video generation techniques

Name	Methodology	Technology/ Framework	Outcome	Strengths	Limitations
CogVideo	Multi-frame-rate hierarchical training	Fine-tuning a pre-trained text-to-image model	Improved text-clip alignment and accuracy	Efficient fine-tuning and control over intensity	Text-action relation challenges, dataset issues
C3LLM	Multimodal LLM with hierarchical audio generation	LLM backbone, hierarchical tokenisation	Versatile multimodal generation	Unified model, improved semantic alignment	Complexity in training and real-time challenges
Make-a-Video	Generative adversarial networks (GANs)	GANs with advanced video synthesis techniques	High-quality video content	High-quality visuals and diverse input handling	High computational resources and large datasets
Mo, Shi, and Tian (2024)	Text-to-audio generation synchronised with video	Synchronisation algorithms	Coherent and synchronised audio-visual outputs	Effective audio-video synchronisation	Real-time synchronisation for complex videos

Multimodal Content Generation Service

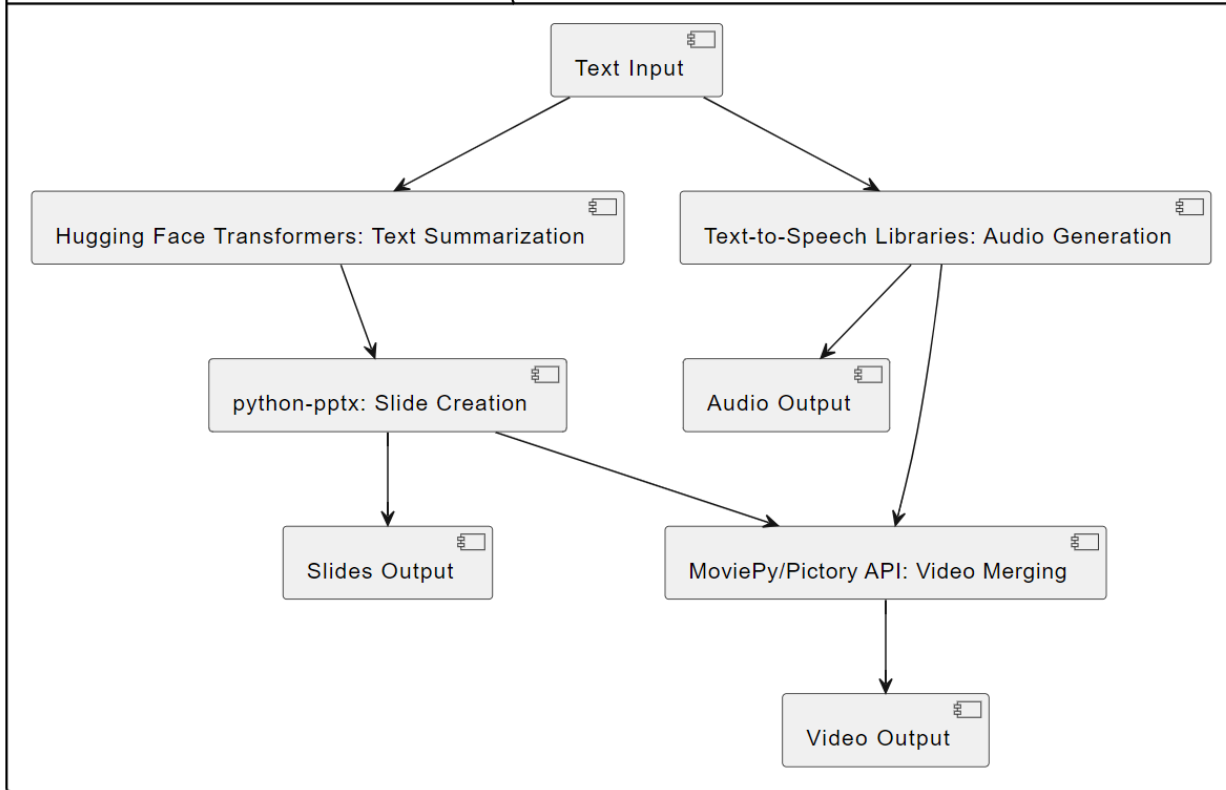


Figure 3: Multimodal content generation service architecture

2.2.2 Recommender System

The reviewed recommender systems employ various techniques for personalised learning. Imran et al.'s PLORS system uses a hybrid approach combining collaborative filtering and content-based methods with ontology-based mapping for semantic understanding. Bhaskaran and Marappan's system utilises advanced machine learning techniques, including clustering and sequential pattern mining, to analyse user data, focusing on scalability. Idrissi et al. (2023) integrate collaborative filtering, content-based methods, and autoencoders for dynamic and personalised recommendations. Aher and Lobo's approach employs simple k-means clustering and the association rule algorithm to identify effective learning sequences and group similar learners. Hybrid approaches, balancing multiple techniques, are particularly effective, offering improved recommendation accuracy and personalisation. Given the strengths of hybrid techniques in enhancing recommendation accuracy and personalisation, adopting a hybrid approach is recommended for the AdaptEd plugin. This will integrate collaborative filtering and content-based methods, combining their strengths to provide more accurate and personalised course recommendations.

Table 6: Comparative analysis of recommender systems

Sources	Primary Techniques	Additional Techniques	Strengths	Weaknesses	Complexity
PLORS (Imran et al., 2016)	Hybrid (Collaborative + Content-Base)	Ontology-Based Mapping	Enhanced semantic matching	Dependence on comprehensive profiles/ontologies	Moderate to High
Bhaskaran and Marappan (2023)	Machine Learning (Clustering, Sequential Mining)	Feature Extraction, Scalability	Advanced analysis, high scalability	Requires extensive data and resources	High
Idrissi et al. (2023)	Hybrid (Collaborative + Content-Base)	Autoencoders	Dynamic adaptation, high relevance	Complexity in integrating autoencoders	High
Aher and Lobo (2011)	Clustering (K-Means), Association Rule Algorithm	Collaborative Filtering	Insights from learning sequences and effective grouping	Dependency on clustering accuracy	Moderate

2.2.3 Learning Analytics and Early Prediction of At-Risk Students

Learning analytics and data mining techniques monitor student performance and predict at-risk students using machine learning, data mining algorithms, and predictive modelling. Three key studies on predicting at-risk students implemented various machine learning techniques. Abdrakhmanov et al. (2024) used Convolutional Neural Networks (CNN) to predict academic performance in STEM and non-STEM education, offering high accuracy, robust feature extraction, and scalability. Alalawi et al. (2021) used RF, SVM, DT, KNN, and NB, achieving the highest binary classification accuracy of 96% with KNN, and introduced a web application for real-time prediction and intervention. Chui et al. (2020) focused on a variant of SVM, Reduced Training Vector Support Vector Machine (RTV-SVM), which reduced training time while maintaining high accuracy and performing well in both binary and multiclass predictions.

Overall, ensemble methods like RF and AdaBoost showed high accuracy but had higher computational demands, making SVM variants, especially RTV-SVM, more suitable for large-scale and real-time applications.

Recent studies suggest using ensemble and hybrid models for educational data to improve accuracy by combining different algorithms (Abdrakhmanov et al., 2024). Therefore, AdaptEd will implement RTV-SVM and RF, leveraging the strengths of both algorithms for enhanced accuracy and robustness.

Key Studies on Predicting At-Risk Students

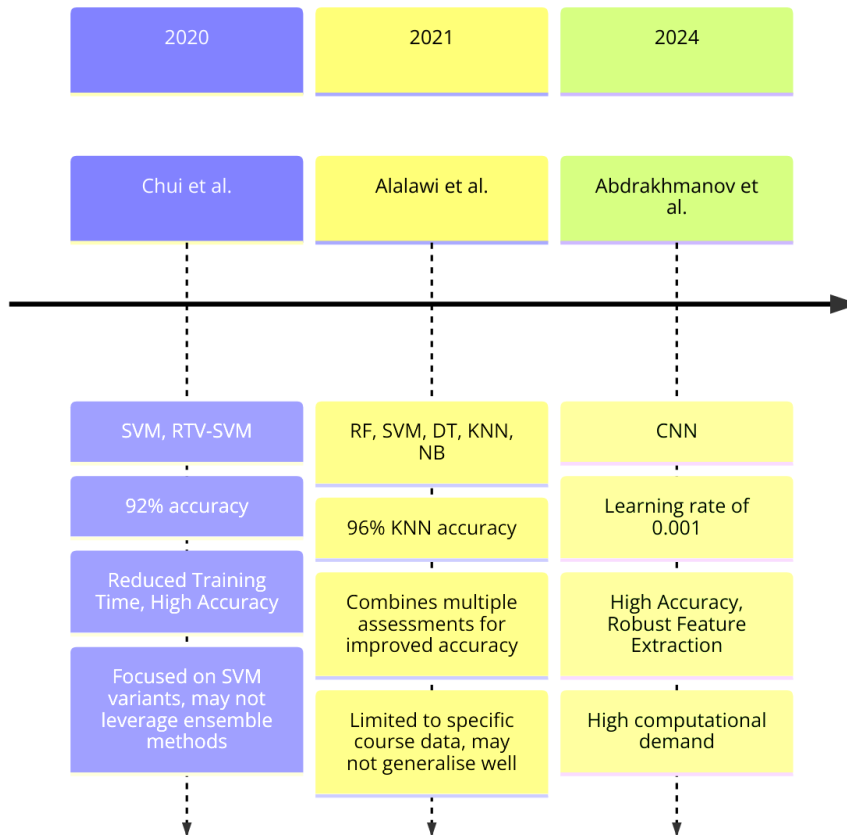


Figure 4: Techniques for early prediction of at-risk students

2.2.4 AI-Powered Virtual Assistant

AI and NLP-powered virtual assistants provide 24/7 support and enhance student engagement. Radziwill et al. (2017) explored AI-powered virtual assistants, demonstrating advanced contextual understanding and personalised support, but with complex development. Cahn (2017) examined NLP-based chatbots, effectively responding to a wide range of queries but struggling with complex queries and requiring continuous updates.

Radziwill et al.'s (2017) AI-powered virtual assistant approach is recommended for AdaptEd due to its advanced contextual understanding and personalised support.

Table 7: comparative analysis of virtual assistants

Sources	Approach	Outcome	Strengths	Limitations
Radziwill, Nicole M., et al. (2017)	AI-Powered Virtual Assistants	Advanced contextual understanding and personalised support	High engagement and contextual understanding	Complex to develop and maintain requires significant resources.
Cahn (2017)	NLP-Based Chatbots	Understands and responds to a wide range of queries	Improves student support and engagement	May struggle with complex queries and require continuous updates.

3.0 Methods

3.1 System Design Analysis

3.1.1 Architecture Overview

Microservices Architecture

AdaptEd is designed using a microservices architecture, an architectural style that structures an application as a collection of loosely coupled services, each responsible for a specific functionality. Microservices are independently deployable and scalable, which aligns well with the needs of modern, complex applications like AdaptEd that require flexibility, scalability, and maintainability (Dragoni et al., 2017). Each feature of the tool—multimodal content generation, recommendation system, learning analytics, and virtual assistant—will be developed as an independent service that communicates with others through APIs. In comparison, a monolithic architecture integrates all functionalities into a single codebase, simplifying development and deployment initially but becoming unwieldy and difficult to manage as the application grows, with updates and scaling becoming challenging (Balalaie et al., 2016). Service-Oriented Architecture (SOA) is similar to microservices in decomposing applications into services, but SOA services are typically larger and share resources more than microservices. Microservices offer greater independence and finer granularity (Lewis & Fowler, 2014). Figure 5 below shows the architecture diagram of AdaptEd.

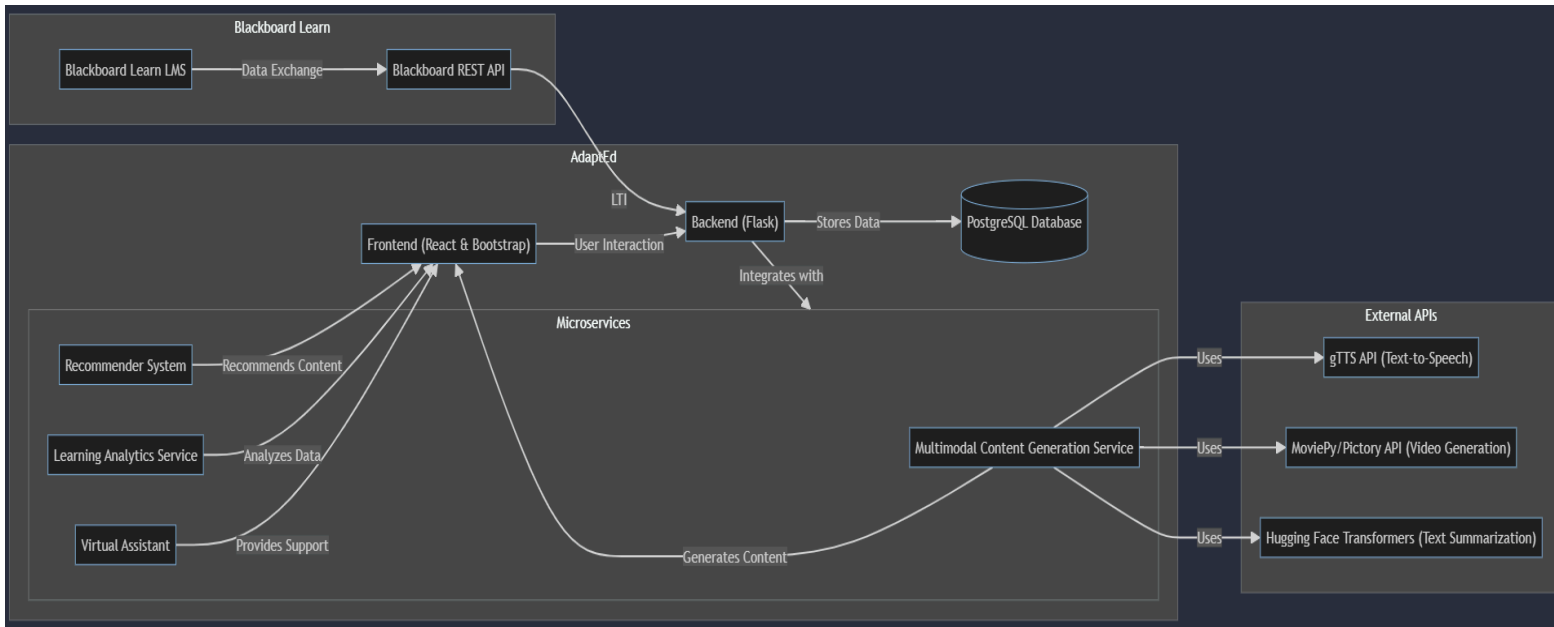


Figure 5: AdaptEd's system architecture

Justification for Microservices Architecture

Table 8: Justification for microservices architecture

Reason	Justification
Scalability	Each service can be scaled independently based on demand, ensuring efficient resource use (Newman, 2015).
Flexibility	Allows for the use of different technologies for different services, enhancing flexibility (Lewis & Fowler, 2014).
Fault Isolation	Issues in one service do not directly impact others, improving overall system resilience (Dragoni et al., 2017).
Continuous Deployment	Facilitates continuous deployment and integration, crucial for Agile development (Newman, 2015).

3.2 Tools and Technology

The development of AdaptEd utilises a comprehensive tech stack comprising various libraries, frameworks, and services. Python serves as the primary programming language, supported by Flask for the web framework. React and Bootstrap are employed for the user interface to ensure a responsive and interactive design. The virtual assistant functionality is implemented using the ChatGPT API, while TensorFlow Recommenders and Scikit-Learn power the recommendation system. Text summarisation for slide generation is managed by Hugging Face Transformers, with Python-pptx employed for the

creation of slides. Video generation is achieved through the integration of text-to-speech libraries with MoviePy or the Pictory API. For early prediction of at-risk students, the stack incorporates PyTorch for model building and training, alongside Scikit-Learn for data pre-processing and traditional machine learning algorithms. PostgreSQL is used for data storage and management, with Git facilitating version control. The tool will be deployed on AWS. Table 9 below shows the justification for the tools and technologies.

Table 9: Justification for tools and technologies

Component	Tool/Technology	Alternative	Justification
Web Framework	Flask	Django	Flask is lightweight, flexible, and widely used for building microservices. Its simplicity allows for quick development and easy maintenance (Grinberg, 2018).
Frontend	React & Bootstrap	Angular	React offers component-based architecture, facilitating reusable UI components. Bootstrap ensures responsive design, enhancing user experience across devices (Banks & Porcello, 2020; Otwell, 2018).
Database	PostgreSQL	MySQL	PostgreSQL is a robust, open-source relational database known for its reliability and support for complex queries (Momjian, 2001).
Deployment	AWS	Azure	AWS provides scalable and reliable cloud services, essential for deploying and managing microservice architecture (Amazon Web Services, 2024).
Recommendation System	TensorFlow Recommenders	Apache Mahout	TensorFlow Recommenders leverages powerful machine learning models to deliver accurate recommendations based on user data (Chollet, 2017).
Learning Analytics	PyTorch	TensorFlow	PyTorch is favoured for its dynamic computation graph and ease of use in developing and testing machine learning models (Paszke et al., 2019).

Data Pre-processing	Scikit-Learn	Pandas	Scikit-Learn is a comprehensive library for data mining and analysis, providing efficient tools for data preprocessing (Pedregosa et al., 2011).
Text Summarisation	Hugging Face Transformers	SpaCy	These pre-trained models provide state-of-the-art performance for NLP tasks such as text summarisation (Wolf et al., 2020).
Slide Generation	python-pptx	LibreOffice API	Python-pptx allows for the programmatic creation and manipulation of PowerPoint files, facilitating automated slide generation (Taylor, 2020).
Text-to-Speech	gTTS	Amazon Polly	gTTS is a simple yet effective tool for converting text to speech using Google's Text-to-Speech API (Google Text-to-Speech, 2024).
Video Generation	MoviePy/Pictory API	OpenShot API	MoviePy is a flexible library for video editing, while Pictory API offers automated video creation suitable for educational content (Pictory AI, 2024; Zulko, 2024).
Version Control	GitHub	GitLab	GitHub provides robust version control, collaboration features, and CI/CD integration, essential for managing development projects (GitHub, 2024).
Integration	LTI & Blackboard REST API	Custom API	LTI ensures interoperability with LMS, and Blackboard REST API allows seamless integration with Blackboard Learn (IMS Global, 2020; Blackboard, 2024).

3.3 Agile Development Methodology

AdaptEd will be developed using Agile methodology to facilitate iterative development, continuous feedback, and rapid adaptation to changes. This approach ensures the tool evolves based on user feedback and changing requirements (Beck et al., 2013). In comparison with other software development methodologies, the waterfall model follows a linear and sequential approach, making it rigid and difficult to accommodate changes once a phase is completed (Royce, 1970). This inflexibility is unsuitable for projects like AdaptEd, which require iterative development and continuous integration. The V-Model, an extension of the waterfall model, emphasises verification and validation but shares the

same rigidity and lack of support for iterative development (Forsberg & Mooz, 1991). The spiral model combines iterative development with systematic aspects of the waterfall model, making it more flexible than the waterfall approach. However, it can be complex and costly due to its iterative nature and the requirement for risk analysis at each phase (Boehm, 1988). Agile methodologies strike a balance between flexibility and structure, allowing for iterative development while providing a framework for managing the process effectively.

Justification for Agile Methodology

Table 10: Justification for Agile methodology

Reason	Justification
Flexibility and Adaptability	Agile allows for changes and refinements throughout the development process, which is essential for accommodating evolving user requirements (Schwaber & Beedle, 2002).
Customer Collaboration	Regular feedback from stakeholders ensures the product meets user needs and expectations (Beck et al., 2001).
Incremental Delivery	Delivers functional components in iterations, providing value to users early and often (Schwaber & Beedle, 2002).
Risk Management	Frequent testing and review reduce the risk of significant issues going unnoticed until later stages (Sutherland, 2014).

3.3.1 Agile Phases and Practices

Scrum

Scrum is a popular Agile framework that structures development in cycles of work called sprints, typically lasting two weeks. Each sprint aims to produce a potentially shippable product increment (Schwaber & Sutherland, 2017).

Key Practices:

- **Sprint Planning:** Define the scope and objectives of the sprint, breaking down tasks and assigning them to team members (Sutherland, 2014).
- **Daily Stand-ups:** Hold daily reviews to discuss progress, identify roadblocks, and adjust plans as needed.
- **Development and Testing:** Develop features incrementally, continuously integrating and testing new code.
- **Sprint Review:** Demonstrate completed work to stakeholders, gather feedback, and plan for the next sprint.
- **Sprint Retrospective:** Reflect on the sprint to identify what went well, what didn't, and how processes can be improved.

Implementation Timeline

The timeline for the project is six weeks; hence, the project will be divided into three two-week sprints:

Table 11: Project timeline

Week	Phase	Activity Category	Activities
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1-2	Sprint 1	Planning and Requirement Analysis Initial design Backend Development Frontend Development	Gather detailed requirements from stakeholders, define user stories, and create the product backlog. Develop system architecture and microservice design. Create UI/UX prototypes using Figma. Implement core backend services using Flask and Python. Set up the PostgreSQL database. Develop initial UI components with React and Bootstrap.
3-4	Sprint 2	AI Model Integration API Development Multimodal Content Generation Frontend Enhancement	Integrate TensorFlow Recommenders for the recommendation system and Hugging Face Transformers for text summarisation. Develop APIs for communication between microservices. Refine UI components and integrate with backend services.
5-6	Sprint 3	Advanced Backend Development Frontend Enhancement Testing and Deployment	Implement learning analytics using PyTorch and data pre-processing with Scikit-Learn. Implement the ChatGPT API for the virtual assistant. Perform integration testing, user acceptance testing (UAT), and security testing. Deploy the tool on AWS and conduct final performance monitoring.

3.4 Data Collection and Analysis

3.4.1 Data Collection

Surveys: Administer questionnaires to educators and students to gather comprehensive insights into their experiences, preferences, and challenges with existing learning systems. The survey will probe into aspects such as preferred learning modalities, the effectiveness of current LMS tools, and feedback on adaptive learning technologies.

System Logs and Usage Data: Collect detailed data on student interactions with the LMS, including metrics such as time spent on various content types, engagement levels, and performance outcomes. This data will be critical for understanding user behaviour and system efficacy.

3.4.2 Data Analysis

Descriptive Statistics: Employ descriptive statistical methods to summarise the survey results, identifying prevalent trends and common user preferences. This analysis will highlight key areas for improvement and user satisfaction.

Predictive Analytics: Utilise machine learning models, specifically PyTorch, to predict at-risk students based on engagement and performance data. Feature engineering techniques in Scikit-Learn will be applied to preprocess the data, enhancing model accuracy and predictive power.

3.5 Evaluation and Assessment

The evaluation and assessment of the AdaptEd plugin will employ a mixed-methods approach, combining quantitative and qualitative data to provide a comprehensive analysis of its effectiveness. This approach aligns with the principles outlined in the literature on personalised learning evaluation.

Quantitative Methods:

- **Surveys and Questionnaires:** Administered to students and educators to gather data on user satisfaction, perceived effectiveness, and usability of the AdaptEd plugin.
- **System Analytics:** Collection of usage data such as time spent on the platform, engagement levels, and academic performance metrics to measure the impact on learning outcomes.

Qualitative Methods:

- **Interviews and Focus Groups:** Conducted with a sample of students and educators to obtain in-depth feedback on their experiences and suggestions for improvement.
- **Case Studies:** Detailed examination of individual or group experiences with the AdaptEd plugin to understand its impact on specific learning contexts.

Data from these methods will be analysed to identify trends, strengths, and areas for improvement. The mixed-method approach ensures a robust evaluation by capturing both numerical data and personal insights, providing a holistic view of the plugin's effectiveness in enhancing personalised learning.

3.6 Ethical Considerations

Data Privacy: Ensure stringent adherence to data protection regulations, such as GDPR. Obtain informed consent from all participants regarding data collection and usage (European Union, 2016).

Bias and Fairness: Mitigate potential biases in AI models by employing diverse and representative datasets. Conduct regular audits of the models to ensure they provide fair and equitable treatment to all students (Mehrabi et al., 2021).

Transparency: Maintain a high level of transparency in data usage and AI-driven decision-making processes. Provide users with clear explanations about the functionality and limitations of the system to foster trust and understanding (Doshi-Velez & Kim, 2017).

User Consent: Clearly communicate to users about the types of data being collected and their intended uses. Ensure that users have the option to opt-out at any stage of the process (Nissenbaum, 2011).

Security: Implement robust security measures to safeguard user data against unauthorised access and potential breaches, thereby ensuring the integrity and confidentiality of the data (Sullivan & Liu, 2011).

These findings were subsequently refined and prioritised to develop the functional and non-functional requirements. The analysis phase also involved a critical evaluation of project constraints, leading to the strategic decision to exclude the learning analytics feature from the initial release, reserving it for future updates to ensure a focused and timely delivery of core functionalities.

The requirements were further elaborated through the creation of use cases, a use case diagram, and use case forms, providing a visual and narrative representation of system interactions. This approach facilitated a deeper understanding of user interactions with the plugin and helped identify potential edge cases and alternative scenarios.

4.1.1 Functional Requirements

The functional requirements were meticulously defined to encapsulate the core capabilities of the AdaptEd plugin, directly addressing the needs identified in the survey and aligning with the project's objectives. These requirements form the backbone of the plugin's functionality, ensuring a user-centric and feature-rich learning environment.

Table 12: Functional Requirements

ID	Requirements	Description
FRE01	Generate recommendations	The recommender system should be able to generate course content recommendations in student's preferred formats (e.g., text, video, audio) based on the student's interaction history and preferences.
FRE02	View recommendations	Students should be able to view the generated recommendations.
FRE03	Query virtual assistant	Students should be able to query the chatbot about Moodle courses, resources, deadlines, and receive context-aware responses.
FRE04	Generate multimodal content	Instructors should be able to upload text-based content and convert it into audio, slides, and video automatically.
FRE05	Upload generated content to course page	Instructors should be able to upload the generated content to the course page
FRE06	Access generated content	Students should be able to access the generated content

4.1.2 Non-Functional Requirements

The non-functional requirements were carefully crafted to ensure the AdaptEd plugin meets high standards of performance, usability, and reliability. These requirements address critical aspects such as scalability, compatibility, and accessibility, which are essential for the plugin's successful integration and adoption within diverse educational contexts.

Table 13: Non-Functional Requirements

ID	Requirements	Description
NRE01	Performance	The plugin should be optimised for fast response times and minimal latency in delivering content and running models.
NRE02	Scalability	The plugin should be scalable to handle increasing numbers of users and data without performance degradation.

NRE03	Compatibility	The plugin should be compatible with Moodle version 3.9 and above, integrate seamlessly with existing plugins, and function correctly on major web browsers (Chrome, Firefox, Safari, Edge).
NRE04	Usability	The plugin should have a user-friendly interface accessible to users with varying levels of technical expertise.
NRE05	Accessibility	The system should be designed with accessibility standards (e.g., WCAG) to support users with disabilities.
NRE06	Reliability	The system must provide high availability and minimal downtime to ensure continuous accessibility.
NRE07	Maintainability	The codebase should be well-documented and modular to facilitate future updates and maintenance.

4.1.3 Use Cases

The refinement and prioritisation of functional requirements led to the development of comprehensive use cases, as outlined in Table 6.1. These use cases served as the foundation for creating a detailed use case diagram, which visually represented the system's functionality from an end-user perspective. The accompanying use case forms provided in-depth descriptions of both normal and alternative scenarios, facilitating a thorough understanding of potential user interactions and system behaviors.

Table 14: Use Cases

ID	Actor	User Case
USC1	Instructor	Generate multimodal content
USC2	Instructor	Manage multimodal content
USC3	Instructor	Query virtual assistant
	Student	
USC4	Student	Access multimodal content
USC5	Student	View personalised recommendations
USC6	Admin	Manage user access
USC7	Admin	Manage plugin settings

USC 8	Admin	Update and maintain plugin
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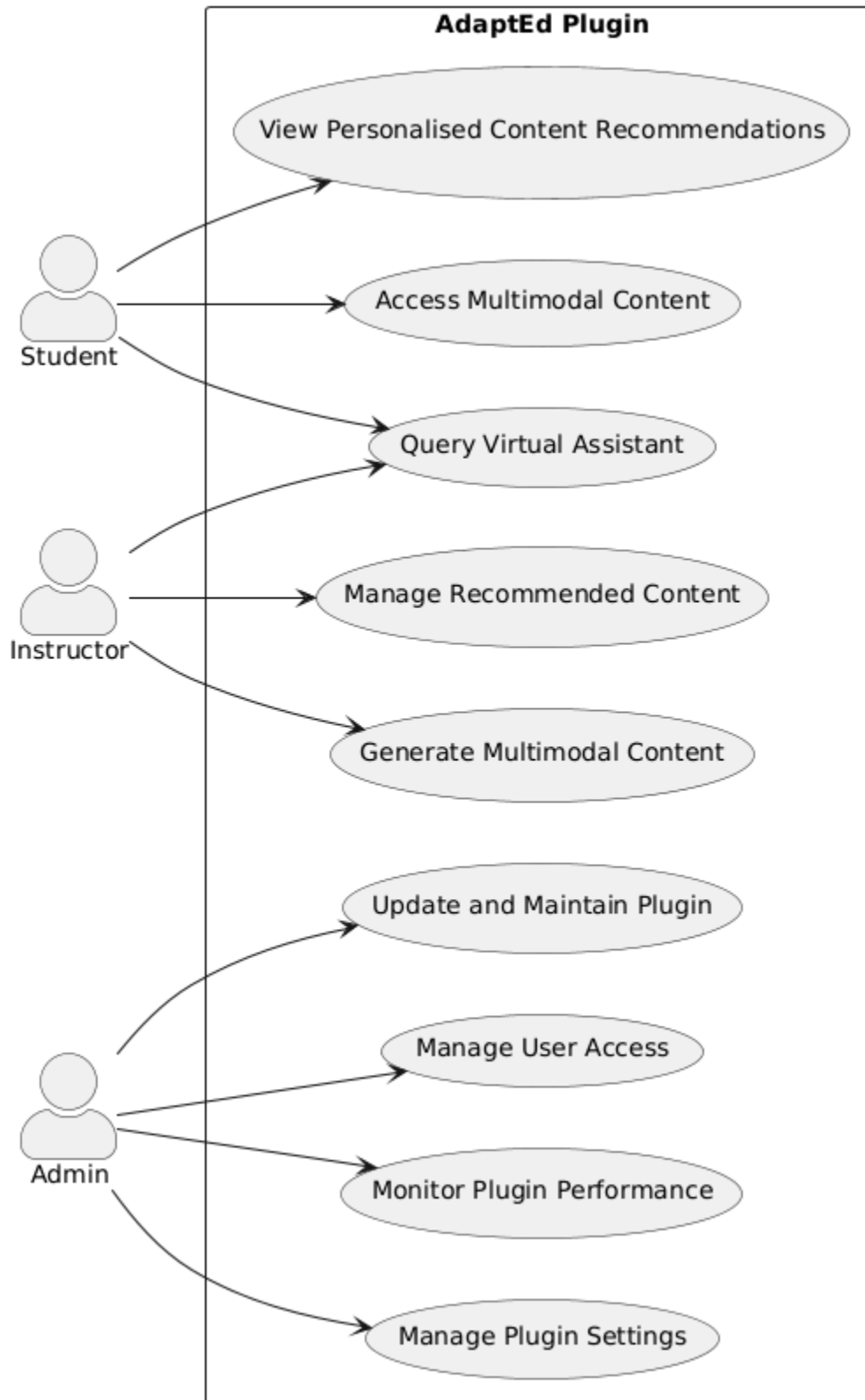


Figure8: Use case Diagram

Use Case Forms

Table 15: Use case: Query Virtual Assistant

Use Case Name	Query Virtual Assistant
Actors	Student, Instructor
Brief Description	Student or Instructor interacts with the chatbot to ask questions about course content, grades, assignments, or other topics.
Pre-Conditions	User is logged into the Moodle system and has access to the AdaptEd plugin.
Possible Scenarios for "Query Virtual Assistant"	
Scenario 1 (Happy Scenario):	
Step	Action
1	User navigates to the AdaptEd chatbot interface
2	User types a question into the chatbot input field
3	System processes the query and retrieves relevant information
4	Chatbot displays a response to the user's query [A1]
5	User reviews the response and can ask follow-up questions if needed
Alternative Scenarios	
Scenario 2 (Sad Scenario):	
Step	Action

A1	If the chatbot cannot understand or find an answer to the query, it prompts the user to rephrase the question or enter valid prompt. Return to step 2 of the happy scenario.
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Table 16: Use case: View Personalised Content Recommendations

Use Case Name	View Personalised Content Recommendations	
Actors	Student	
Brief Description	Student receives and views recommendations for course content in their preferred format (audio, video, text) based on their interaction history.	
Pre-Conditions	Student is logged into the Moodle system and has interacted with course content previously.	
Possible Scenarios for "View Personalised Content Recommendations"		
Scenario 1 (Happy Scenario):		
Step	Action	
1	Student navigates to the course page	
2	System analyses the student's interaction history and preferences [A1]	
3	System generates a list of 10 personalised content recommendations and displays the top three recommendations	
4	Student selects a recommended content item to access	
5	System loads the selected content for the student to interact with	
Alternative Scenarios		
Scenario 2 (Sad Scenario):		
Step	Action	

A1	If user is new or there's not enough interaction history to generate personalised recommendations, the system displays recommendations based on the content. Return to step 4 of the happy scenario.
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Table 17: Use case: Generate Multimodal Content

Use Case Name	Generate Multimodal Content
Actors	Instructor
Brief Description	Instructor uploads text-based content and converts it into audio, slides, and video formats for students to access.
Pre-Conditions	Instructor is logged into the Moodle system and has permission to create and edit course content.
Possible Scenarios for "View Personalised Content Recommendations"	
Scenario 1 (Happy Scenario):	
Step	Action
1	Instructor navigates to the content creation section of the AdaptEd plugin
2	Instructor uploads text-based content [A1]
3	Instructor selects desired output formats (audio, slides, video) [A2]
4	System processes the content and generates the selected formats [A3]
5	System displays a preview of the generated content
6	Instructor reviews and approves the generated content [A4]
7	System uploads the approved content to the course page
Alternative Scenarios	
Scenario 2 (Sad Scenario):	

Step	Action
A1	If the user uploaded a non text-based file, the system displays “Invalid file format” and prompt user to upload a valid file. Return to step 2 of the happy scenario.
A2	If the user did not select at least one output format, the system prompts user to select a format. Return to step 3 of the happy scenario.
A3	If the system encounters technical error during content generation, it displays an error message and suggests troubleshooting steps. Return to step 2 or 3 of the happy scenario.
A4	If the instructor is not satisfied with the generated content, they can edit the original text or adjust generation settings. Return to step 4 of the happy scenario.

This methodical approach to requirements gathering and analysis not only ensured a clear roadmap for development but also established a solid foundation for the subsequent design and implementation phases. By grounding the development process in empirical data and user-centric design principles, the AdaptEd plugin is positioned to effectively address the evolving needs of modern educational environments, fostering enhanced learning experiences through innovative technology integration.

4.2 Design and Modelling

The design and modelling phase of the AdaptEd plugin development involved translating the gathered requirements into a concrete, implementable structure. This phase employed a systematic approach, leveraging modern software design principles and methodologies to ensure the creation of a robust, scalable, and maintainable solution. The design process was guided by the project objectives, particularly the need for seamless integration with Moodle, enhanced user experience, and the provision of adaptive learning features.

4.2.1 System Design

The system architecture for the AdaptEd plugin underwent a significant evolution during the design phase, primarily driven by practical considerations and the need for optimal performance and cost-effectiveness. Initially, the integration was planned for Blackboard Learn; however, a critical analysis of the associated AWS resource costs led to a strategic pivot towards Moodle as the target LMS. This decision was the result of a thorough cost-benefit analysis, which revealed that Moodle offered a more sustainable and cost-effective platform for the plugin's deployment. The open-source nature of Moodle aligns well with the project's objectives of accessibility and scalability, allowing for greater flexibility in customization and integration.

The revised system architectural design, as illustrated in the image below, showcases a microservices-based approach. This architecture was chosen for its ability to provide modular, independently deployable services that can be developed, updated, and scaled individually. The key components of the architecture are highlighted in the table below.

Table 18: System Components

Components	Description and Justification
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Frontend • Mustache • NGINX	Mustache was used for templating in place of React and Bootstrap that were intended for the initial Blackboard Learn because of its usability in Moodle development and easy integration as it is Moodle's native template system. NGINX was used as a reverse proxy to ensure efficient request handling and load balancing.
Backend • Flask	Flask is a lightweight Python web framework that allows for rapid development and easy integration with various services.
Microservices • Recommender System • Multimodal Content Generator • Chatbot Service	The plugin features are broken down to microservices for flexibility and scalability. Moodle is written in PHP and some backend processes like generating recommendations, are largely dependent on Python, separating the features will enable the use of different technologies and programming languages for different services based on their specific requirements.
Database • PostgreSQL	PostgreSQL is used for persistent data storage, chosen for its robustness and support for complex queries.
External Services • gTTS • Python-pptx • Hugging Face Transformers • Moviepy • OpenAI API • OpenCV • DALL-E	These external services along with other libraries that will be discussed in the implementation section are used to process various backend tasks such as text conversion, model training and recommendation generation, natural language processing tasks, and advanced language model capabilities.

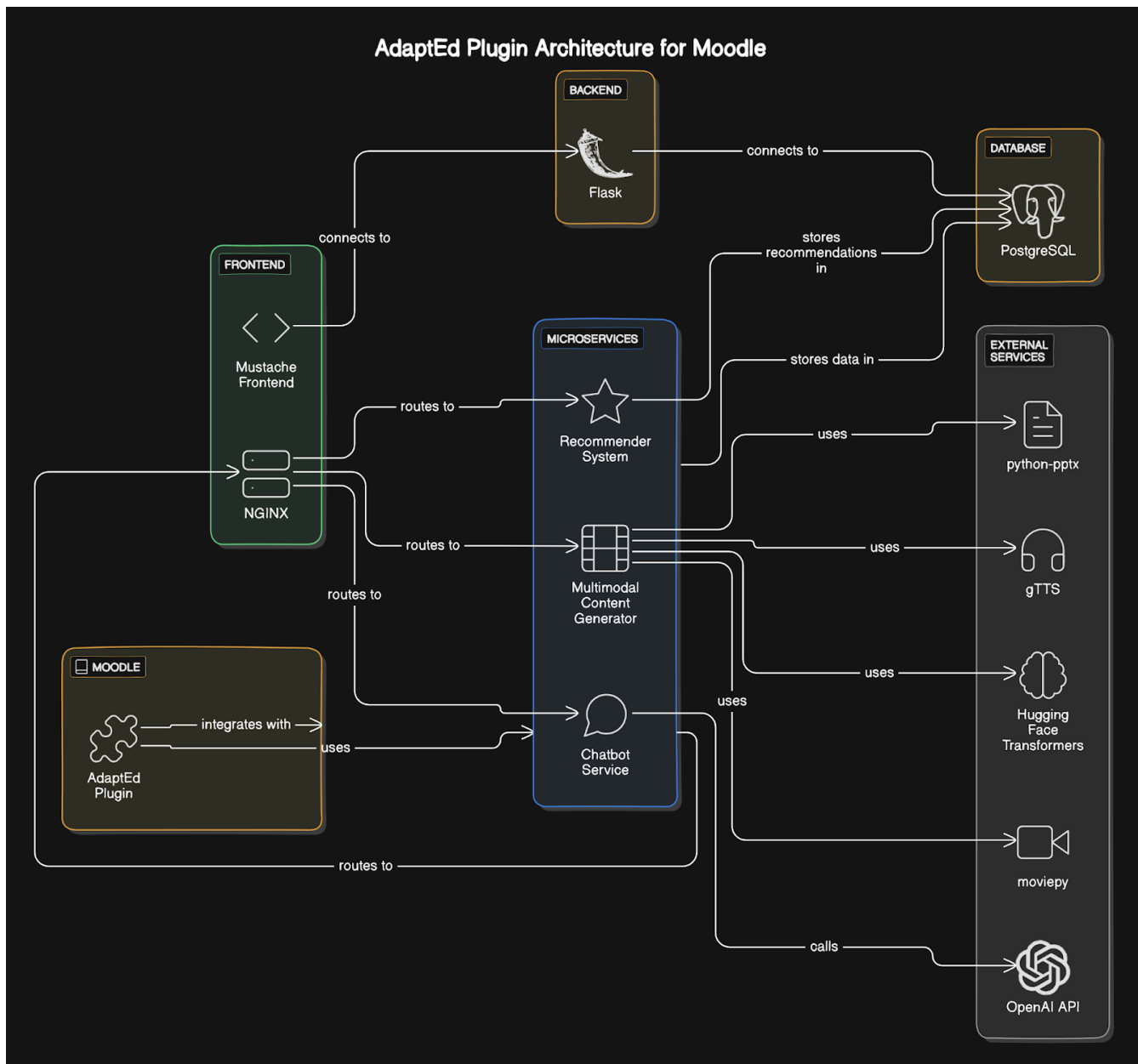


Figure 9: System Architecture design

The design also incorporates principles of loose coupling and high cohesion, ensuring that each component has a well-defined responsibility while minimizing dependencies between services. This approach facilitates easier maintenance, testing, and future enhancements of the plugin.